

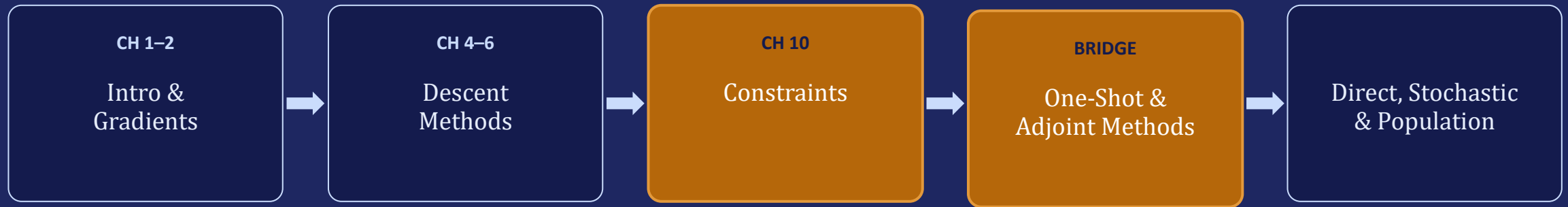
Constrained Optimization

Lecture 3 — Lagrange Multipliers, KKT Conditions, and Penalty, Augmented Lagrange & Interior Point Methods

- Lagrange multipliers & KKT
- Penalty & interior point methods
- One-shot / adjoint methods

Course Roadmap

Lectures 1–2 assumed x could move freely. Today's chapter — and its closing bridge to optimal control — handles the case where x must satisfy constraints.



Today's lecture

Chapters 7–9 (direct, stochastic, population methods) remain from the original Algorithms for Optimization roadmap. Today also previews how these ideas scale to PDE-constrained optimal control problems.

CH 7–9

PART A · SECTIONS 10.1–10.3

Fundamentals

Constraint types, and how constraints reshape the problem

10.1 CONSTRAINED OPTIMIZATION

Recalling the Feasible Set

Every problem in this course has had this same shape:

$$\text{minimize}_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to } \mathbf{x} \in \mathcal{X}$$

Unconstrained problems

The feasible set $\mathcal{X}$ is all of $\mathbb{R}^n$ — $\mathbf{x}$ can be anything.

Constrained problems

The feasible set is some proper subset of $\mathbb{R}^n$ — $\mathbf{x}$ must satisfy conditions.

Constraints arise naturally: a hedge fund can't sell more stock than it owns, an airplane wing can't have zero thickness. We introduce constraints to keep the optimizer from suggesting infeasible solutions.

10.2 CONSTRAINT TYPES

Equality and Inequality Constraints

Feasible sets are typically built from two kinds of constraint functions:

$$h(\mathbf{x}) = 0 \qquad g(\mathbf{x}) \leq 0$$

Equality constraint

$$\mathbf{h}(\mathbf{x}) = 0$$

Design point must lie exactly on this surface.

Inequality constraint

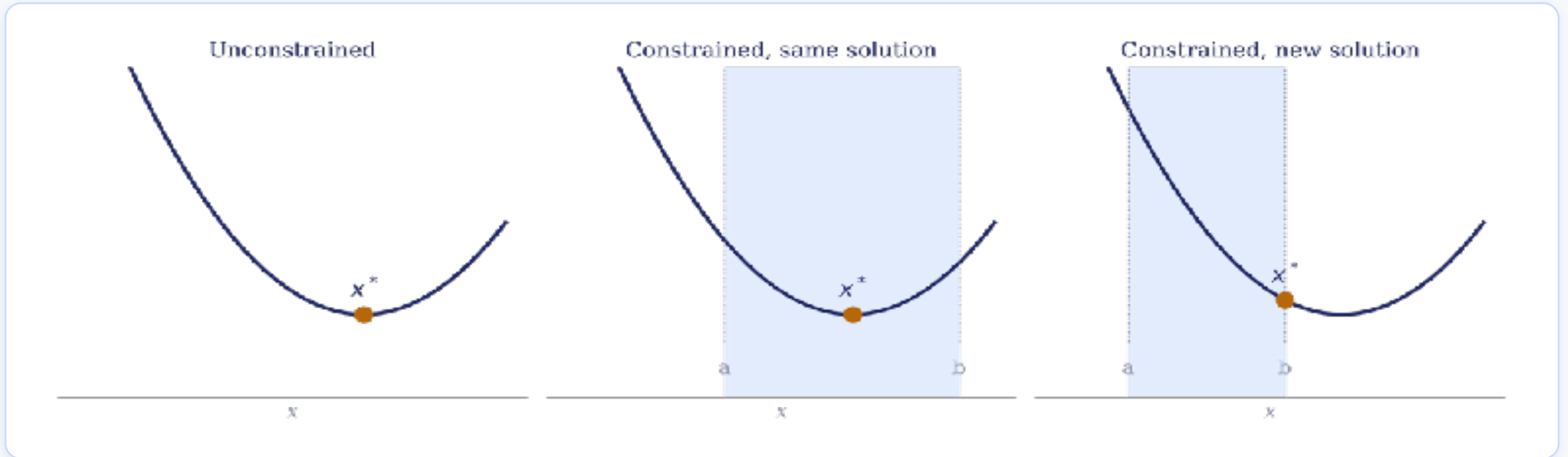
$$\mathbf{g}(\mathbf{x}) \leq 0$$

Design point must lie on one side of this surface.

Any constrained problem can be written using these two building blocks — equality constraints $h_i(\mathbf{x})=0$ and inequality constraints $g_j(\mathbf{x})\leq 0$ for as many i and j as needed.

10.2 CONSTRAINT TYPES

Constraints May or May Not Change the Answer



If the unconstrained minimum already lies inside the feasible region, adding the constraint changes nothing. If it lies outside, the constrained solution moves to the nearest feasible point — often right on the boundary.

Sometimes You Can Just Transform the Problem

Bound constraints $a \leq x \leq b$ can be removed entirely by passing x through a smooth transform of an unconstrained variable $\hat{x}$:

$$x = t_{a,b}(\hat{x}) = \frac{b+a}{2} + \frac{b-a}{2} \left(\frac{2\hat{x}}{1+\hat{x}^2} \right)$$

As $\hat{x}$ ranges over all real numbers, $x = t_{a,b}(\hat{x})$ is guaranteed to stay in $[a,b]$. You can then run any unconstrained optimizer on $\hat{x}$ and map the result back through the transform — no constraint-handling logic needed.

Equality constraints can sometimes eliminate a variable directly: if $h(x)=0$ can be solved for x_n in terms of $x_1, \dots, x_{n-1}$, substitute it in and optimize over one fewer variable.

Part A — Key Takeaways

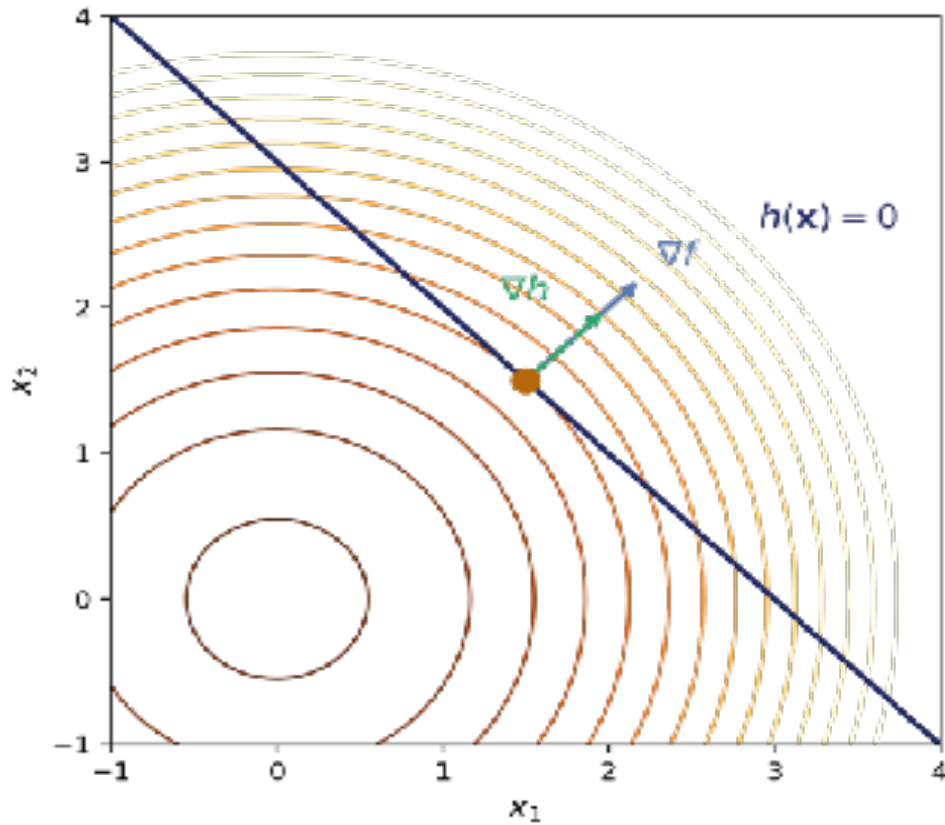
- 1 Constrained problems restrict x to a feasible set built from equality and inequality constraints.
- 2 Constraints can change the optimal solution — or leave it untouched, if it was already feasible.
- 3 Bound constraints can be removed entirely with a smooth transform of an unconstrained variable.
- 4 Some equality constraints let you eliminate a design variable directly, reducing problem dimension.
- 5 When these tricks don't apply, we need genuinely constrained methods — the subject of Part B.

PART B · SECTIONS 10.4–10.6

Lagrange Multipliers & KKT Conditions

The analytical heart of constrained optimization

Where Contour Lines Align



The idea

At the constrained optimum, the contour line of f is tangent to the constraint curve $h(x)=0$. Small moves along h cannot improve f .

Since ∇f and ∇h are both perpendicular to their own contour/constraint lines at that point, tangency means the two gradients must be parallel:

$$\nabla f(\mathbf{x}) = \lambda \nabla h(\mathbf{x})$$

for some scalar λ — the Lagrange multiplier. We need the scalar because the two gradients' magnitudes need not match.

The Lagrangian

We package the gradient-alignment condition and the constraint itself into a single function of $\mathbf{x}$ and λ , the Lagrangian:

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda h(\mathbf{x})$$

Solving $\nabla_{\mathbf{x}, \lambda} \mathcal{L}(\mathbf{x}, \lambda) = 0$ recovers both conditions at once:

$\nabla_{\mathbf{x}} \mathcal{L} = 0$ gives $\nabla f = \lambda \nabla h$ (gradient alignment), and

$\nabla_{\lambda} \mathcal{L} = 0$ gives $h(\mathbf{x}) = 0$ (the constraint itself).

Any solution is a critical point — possibly a local min, global min, or saddle point.

This is a first-order necessary condition for optimality, just like FONC in Chapter 1 — it tells us where to look, not that we've certainly found a minimum.

Generalizing to Multiple Constraints

With ℓ equality constraints $h_1(\mathbf{x})=0, \dots, h_\ell(\mathbf{x})=0$, each gets its own multiplier, giving a single combined Lagrangian:

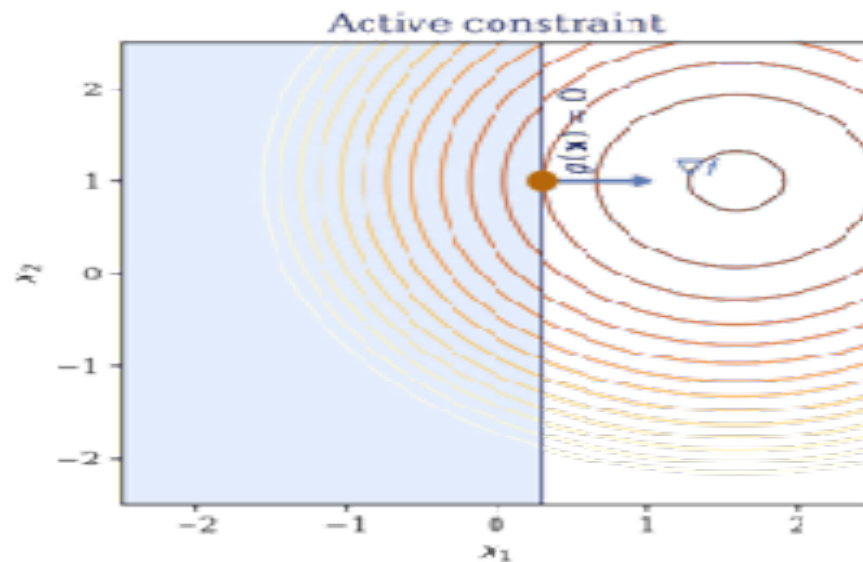
$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{i=1}^{\ell} \lambda_i h_i(\mathbf{x}) = f(\mathbf{x}) - \lambda^\top \mathbf{h}(\mathbf{x})$$

Why not just combine constraints into one, like $h_{\text{comb}} = h_1^2 + h_2^2$? You could — the solution set is identical — but you'd lose the separate multipliers λ_i , which turn out to carry useful information (e.g. sensitivity of the optimum to each constraint).

This vector form $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda^\top \mathbf{h}(\mathbf{x})$ is exactly what generalizes to inequality constraints next.

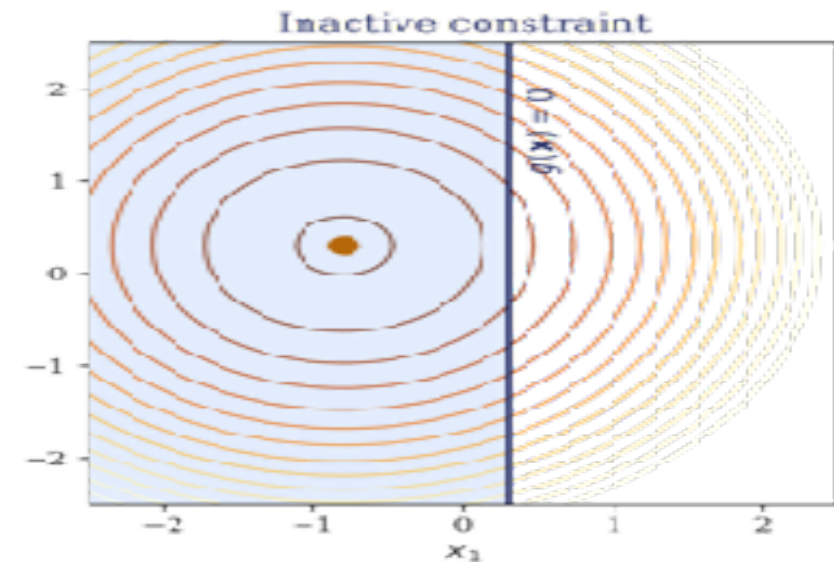
10.5 INEQUALITY CONSTRAINTS

Active and Inactive Constraints



Active: $g(x^*) = 0$

The unconstrained minimum lies outside the feasible region, so the solution sits right on the boundary. Just like an equality constraint, ∇f is limited to be parallel to ∇g .



Inactive: $g(x^*) < 0$

The unconstrained minimum already satisfies the constraint. The solution is a normal unconstrained critical point — the constraint might as well not be there.

10.5 INEQUALITY CONSTRAINTS

A Penalty That Becomes a Lagrangian

For an active constraint, the Lagrange condition holds with some constant μ :

$$\nabla f(\mathbf{x}^*) - \mu \nabla g(\mathbf{x}^*) = 0$$

We can recover an infinite step penalty (∞ if infeasible) by maximizing a linear penalty $\mu g(x)$ over $\mu \geq 0$, then use that to build a Lagrangian for inequality constraints:

$$\mathcal{L}(x, \mu) = f(x) + \mu g(x)$$

The reformulated problem, $\min_x \max_{\mu \geq 0} \mathcal{L}(x, \mu)$, is called the primal problem. For any infeasible x we get infinity; for any feasible x we recover $f(x)$ exactly.

10.5 INEQUALITY CONSTRAINTS

The KKT Conditions

Generalizing to any number of equality and inequality constraints gives four first-order necessary conditions — named after Karush, Kuhn, and Tucker:

1. Feasibility

$$\mathbf{g}(\mathbf{x}^*) \leq 0$$

$$\mathbf{h}(\mathbf{x}^*) = 0$$

All constraints are satisfied.

2. Dual feasibility

$$\boldsymbol{\mu} \geq 0$$

Every inequality multiplier is non-negative.

3. Complementary slackness

$$\boldsymbol{\mu} \odot \mathbf{g} = 0$$

Either $\mu_i = 0$, or $g_i(\mathbf{x}^*) = 0$ — never both nonzero.

4. Stationarity

$$\nabla f(\mathbf{x}^*) - \sum_i \mu_i \nabla g_i(\mathbf{x}^*) - \sum_j \lambda_j \nabla h_j(\mathbf{x}^*) = 0$$

10.6 DUALITY

The Primal and Dual Problems

The full Lagrangian with all multipliers is called the generalized Lagrangian:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_i \mu_i g_i(\mathbf{x}) + \sum_j \lambda_j h_j(\mathbf{x})$$

Primal form

$$\text{minimize}_{\mathbf{x}} \text{ maximize}_{\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}} \mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda})$$

Identical to the original problem — just as hard to optimize.

Dual form

$$\text{maximize}_{\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}} \text{ minimize}_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda})$$

Reverses the order of min and max. Often much easier to solve.

$$\mathcal{D}(\boldsymbol{\mu} \geq 0, \boldsymbol{\lambda}) \leq p^*$$

the dual value is always a lower bound on the primal value — “weak duality.”

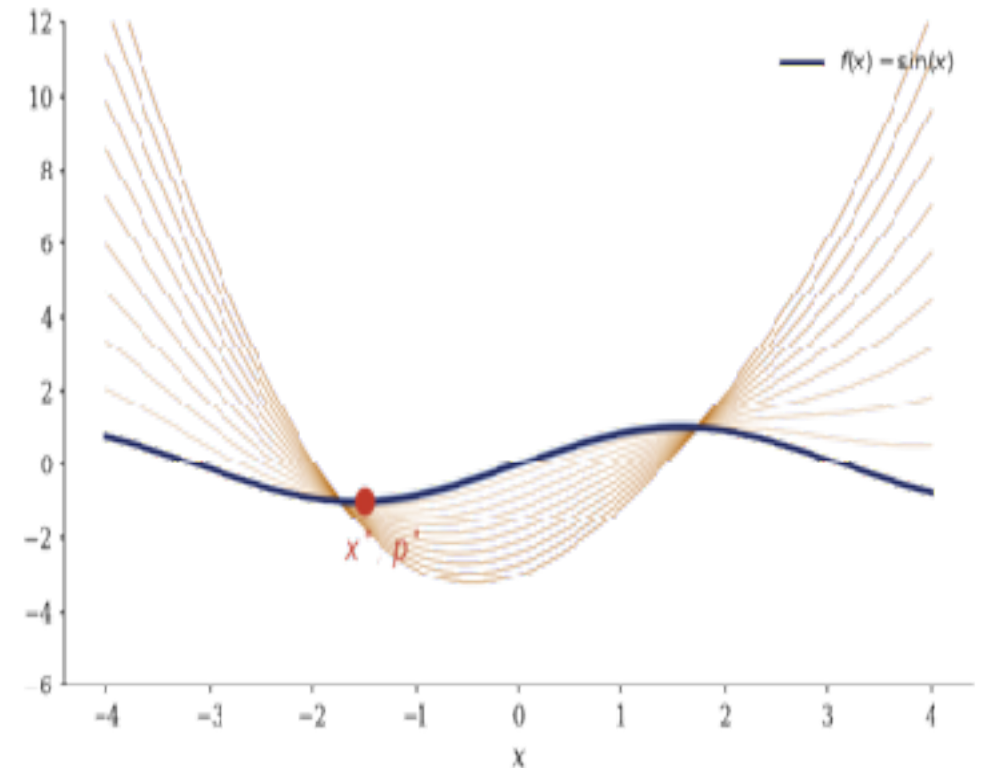
10.6 DUALITY

The Duality Gap

$$\text{duality gap} = p^* - d^*$$

is the difference between the primal and dual optimal values. In some cases — including many convex problems — the gap is zero, and duality gives an alternative way to solve the original problem exactly.

Example: minimize $\sin(x)$ subject to $x^2 \leq 3$. Each choice of $\mu \geq 0$ gives a Lagrangian curve whose unconstrained minimum lies at or below p^* . Maximizing over μ pulls the dual value up to meet p^* from below.



Part B — Key Takeaways

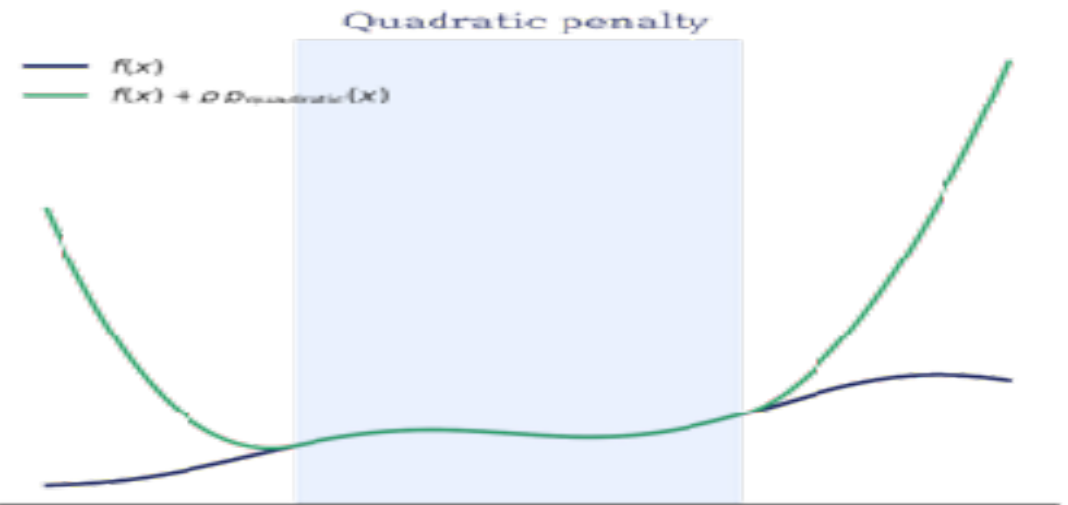
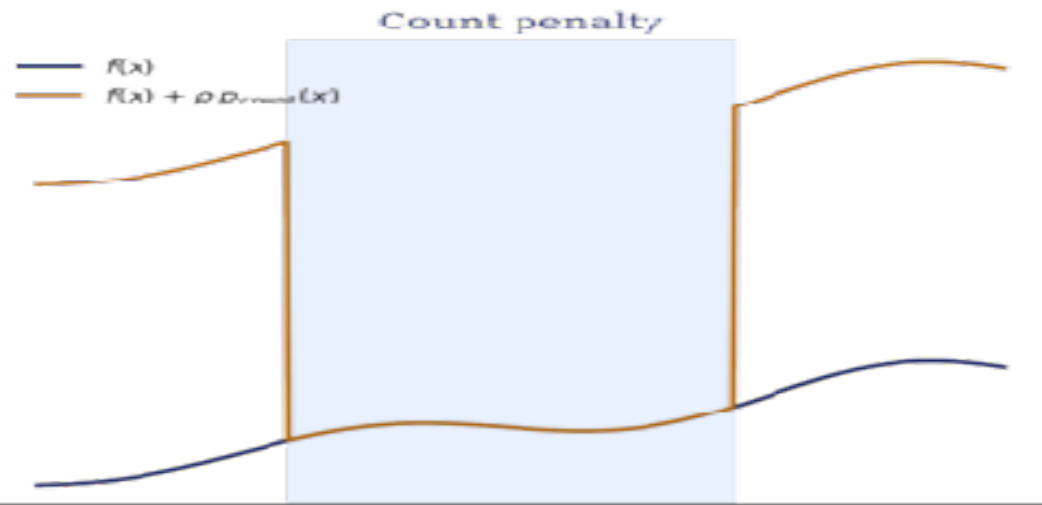
- 1 Lagrange multipliers find where an objective's contour is tangent to an equality constraint.
- 2 The Lagrangian $\mathcal{L} = f - \lambda h$ (or $+ \mu g$) turns a constrained problem into an unconstrained stationarity search.
- 3 For inequalities, a constraint is either active (on the boundary) or inactive (ignorable at the optimum).
- 4 The KKT conditions — feasibility, dual feasibility, complementary slackness, stationarity — generalize FONC.
- 5 Every constrained problem has a dual problem; its value is always a lower bound on the primal's.

10.7 PENALTY METHODS

Penalizing Infeasibility

Penalty methods add a penalty term to the objective, turning the constrained problem back into one our Chapter 4–6 methods can solve directly:

$$\text{minimize}_x f(x) + \rho \cdot p(x)$$



Count, Quadratic, and Mixed Penalties

Count penalty

$$P_{\text{count}}(\mathbf{x}) = \sum_i (g_i(\mathbf{x}) > 0) + \sum_j (h_j(\mathbf{x}) \neq 0)$$

Counts violated constraints — sharp discontinuity, no gradient guidance.

Quadratic penalty

$$P_{\text{quadratic}}(\mathbf{x}) = \sum_i \max(g_i(\mathbf{x}), 0)^2 + \sum_j h_j(\mathbf{x})^2$$

Smooth, but requires $p \rightarrow \infty$ to guarantee feasibility exactly.

Algorithm: start with a small p , solve the unconstrained problem, then increase p and re-solve — using the previous solution as the new starting point — until the result is feasible or a maximum number of iterations is reached.

Combining a Quadratic and a Linear Penalty

The augmented Lagrange method adapts the penalty method for equality constraints, converging without needing $\rho \rightarrow \infty$:

$$p_{\text{Lagrange}}(\mathbf{x}) = \frac{1}{2}\rho \sum_i (h_i(\mathbf{x}))^2 - \sum_i \lambda_i h_i(\mathbf{x})$$

where λ converges toward the true Lagrange multiplier, updated after each iteration alongside ρ :

$$\lambda^{(k+1)} = \lambda^{(k)} - \rho \mathbf{h}(\mathbf{x})$$

Because λ is actively tracking the multiplier rather than relying purely on an ever-growing penalty, the augmented Lagrange method typically needs far smaller values of ρ to reach a feasible, optimal point.

Barrier Functions Keep Us Feasible

Interior point (barrier) methods take a different approach: instead of penalizing infeasibility, they make the boundary itself repulsive, so the search never leaves the feasible region.

Inverse barrier

$$p_{\text{barrier}}(\mathbf{x}) = -\sum_i \frac{1}{g_i(\mathbf{x})}$$

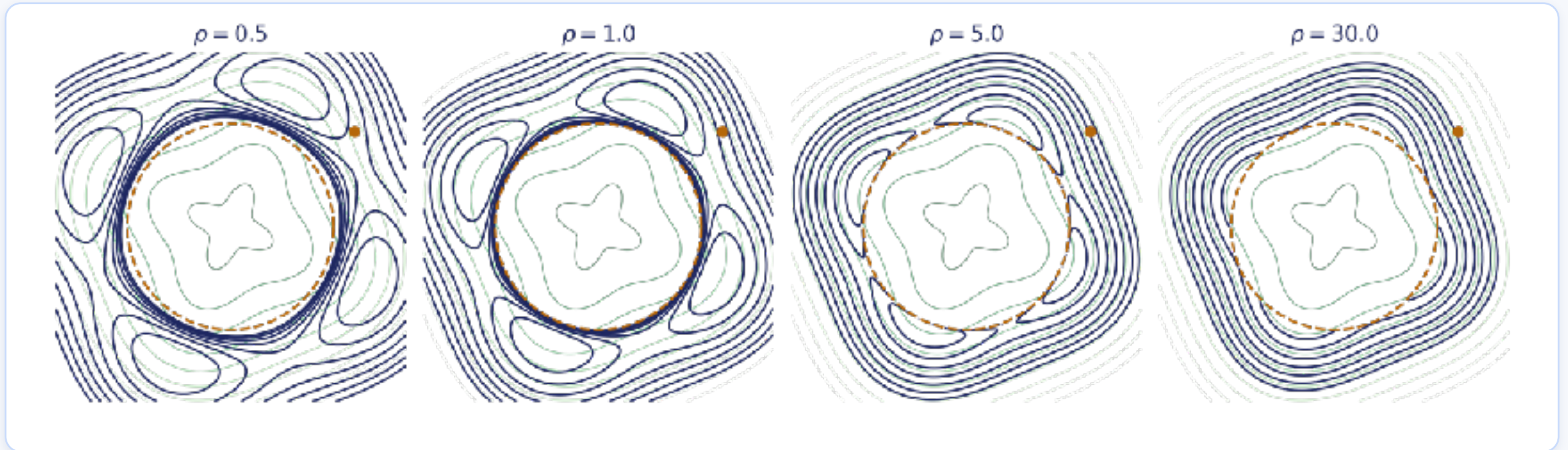
Log barrier

$$p_{\text{barrier}}(\mathbf{x}) = -\sum_i \log(-g_i(\mathbf{x}))$$

$$\text{minimize}_{\mathbf{x}} \quad f(\mathbf{x}) + \frac{1}{\rho} p_{\text{barrier}}(\mathbf{x})$$

As ρ increases, the penalty for approaching the boundary shrinks, letting the optimum approach the true constraint boundary while a valid solution is available at every iteration — useful if you must stop early.

Tracking the Boundary as ρ Increases



The dashed circle marks the true feasible boundary; navy contours are the barrier-augmented objective. As ρ grows, the barrier's influence shrinks and its contours hug the true boundary ever more closely.

The interior point method needs a feasible starting point — often found by first minimizing a quadratic penalty function.

Which Constrained Method, When?

Method	Stays feasible?	Needs $\rho \rightarrow \infty$?	Best for...
Count penalty	No	Yes	Quick, crude feasibility checks
Quadratic penalty	No	Yes (for equality)	Smooth problems, gradient-based solvers
Augmented Lagrange	No	No	Equality constraints, faster convergence
Interior point / barrier	Yes	No ($\rho \rightarrow \infty$ improves precision)	When feasibility must hold at every step

All of these convert a constrained problem into a sequence of unconstrained ones — exactly what our Chapter 4–6 descent methods need.

Part C — Key Takeaways

- 1 Penalty methods add a term $\rho \cdot p(x)$ to the objective, penalizing infeasible points more as ρ grows.
- 2 Count penalties are discontinuous; quadratic penalties are smooth but need $\rho \rightarrow \infty$ for exact feasibility.
- 3 The augmented Lagrange method adds a linear term tracking the true multiplier, converging with smaller ρ .
- 4 Interior point methods use barrier functions to keep every iterate strictly feasible.
- 5 All three reduce a constrained problem to a sequence of unconstrained ones solvable by Chapters 4–6.

PART D · BEYOND THE TEXTBOOK

One-Shot Methods

The same Lagrangian idea, applied to PDE-constrained optimal control

The Same Idea, a Different Setting

Everything in Part B generalizes directly to optimal control and PDE-constrained design problems — just with new names for the same three ingredients:

$\mathbf{x}$ (design point)	$\boldsymbol{\varphi}$ (state variable)	Governed by a physical model — e.g. a velocity field or temperature.
—	$\mathbf{g}$ (control / design parameters)	What we're actually free to choose — e.g. a heat flux or wing shape.
$h(\mathbf{x}) = 0$ (equality constraint)	$\mathbf{F}(\boldsymbol{\varphi}, \mathbf{g}) = 0$ (state equation)	Often a PDE: the Navier–Stokes or heat equations, for example.
λ (Lagrange multiplier)	$\boldsymbol{\xi}$ (adjoint / co-state variable)	Same role: enforces the constraint inside an unconstrained Lagrangian.

State, Control, and the Lagrangian

Let ϕ denote the state, g the control, $\mathcal{J}(\phi, g)$ the objective, and $F(\phi, g)=0$ the governing equation (often a PDE). The optimization problem is:

find g, ϕ such that $\mathcal{J}(\phi, g)$ is minimized subject to $F(\phi, g) = 0$

Exactly as in Part B, we introduce an adjoint variable ξ (the Lagrange multiplier) and form the Lagrangian:

$$\mathcal{L}(\phi, g, \xi) = \mathcal{J}(\phi, g) - \xi^* F(\phi, g)$$

We then seek controls g , states ϕ , and co-states ξ such that $\mathcal{L}(\phi, g, \xi)$ is stationary — the infinite-dimensional analogue of solving $\nabla \mathcal{L} = 0$.

Three Conditions, Just Like KKT

Setting each first variation of $\mathcal{L}$ to zero yields three coupled conditions — the direct analogue of the KKT system:

$$\delta \mathcal{L} / \delta \xi = 0 \quad \frac{\delta \mathcal{L}}{\delta \xi} = 0 \Rightarrow F(\phi, g) = 0$$

Recovers the governing equation $F(\phi, g) = 0$ — just the physics.

$$\delta \mathcal{L} / \delta \phi = 0 \quad \frac{\delta \mathcal{L}}{\delta \phi} = 0 \Rightarrow \left(\frac{\partial F}{\partial \phi} \right)^* \xi - \left(\frac{\partial \mathcal{J}}{\partial \phi} \right)^*$$

The adjoint equation — a linear equation for ξ , given ϕ and g .

$$\delta \mathcal{L} / \delta g = 0 \quad \frac{\delta \mathcal{L}}{\delta g} = 0 \Rightarrow \left(\frac{\partial F}{\partial g} \right)^* \xi - \left(\frac{\partial \mathcal{J}}{\partial g} \right)^*$$

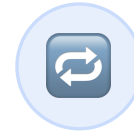
The optimality condition — relates the gradient of $\mathcal{V}$ to ξ .

Solving Everything at Once



The one-shot idea

If the coupled system — state equation, adjoint equation, and optimality condition — can be solved directly, we get the optimal state, co-state, and control simultaneously, with no separate optimization loop.

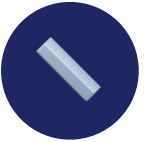


The trade-off

The coupled system is larger and often harder to solve than the state equation alone. For expensive PDE solves (e.g. Navier–Stokes), this can be impractical — motivating iterative alternatives (sensitivity and adjoint-based methods).

The throughline: whether it's a finite-dimensional NLP (Part B) or a PDE-constrained control problem (Part D), the Lagrangian — and its stationarity conditions — is the same idea, just in a different function space.

Constraints, Front to Back



Fundamentals

Constraints define a feasible set X .
Some can be removed entirely with transformations.



Lagrange & KKT

Stationarity of a Lagrangian — and
the four KKT conditions —
characterize constrained optima.



Penalty & Barrier

Penalty, augmented Lagrange, and
interior point methods all reduce
to unconstrained solves.



One-Shot Methods

The same Lagrangian machinery
scales to PDE-constrained optimal
control.

Every method today answers the same question first raised in Chapter 1: how do we characterize — and then find — an optimum, once x is no longer free to go anywhere?

Next Time: Beyond Gradients

Chapters 4–6 and 10 all assumed a gradient (or Lagrangian derivative) was available. Lecture 4 turns to methods that need only function evaluations — or deliberately embrace randomness.

Ch. 7

Direct Methods

Pattern search, Nelder-Mead, DIRECT — gradient-free search using only function evaluations.

Ch. 8

Stochastic Methods

Simulated annealing, cross-entropy method — randomness as a deliberate search tool.

Ch. 9

Population Methods

Genetic algorithms, particle swarm optimization — evolving many candidates together.

Questions?

Reading for next time: Algorithms for Optimization, Chapters 7–9

Kochenderfer, M. J., & Wheeler, T. A. Algorithms for Optimization. MIT Press, 2019.

Part D adapted from Hicken & Zingg-style treatments of one-shot / adjoint methods in PDE-constrained optimization.